VLSI DESIGNING INTERNSHIP

TASK-1

INTRODUCTION

This introduces the basics of logic gates , Half adder and Full adder circuit. The circuits were designed and simulated using Circuitverse to understand their functionality and verifies the outputs using truthtables. And also contains about the VLSI (Very large scale integration) design flow.

VLSI Design Flow

It is the process of integrating millions or billions of transistors on a single silicon chip. And VLSI design flow is a step by step process used to design and manufacture integrated chips. It starts with defining the chip requirements and ends with testing the fabricated chip.

Flow chart

Requirements / Specification

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Architecture Design

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RTL Design (Verilog/VHDL)

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Functional Verification

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Logic Synthesis

↓

Design for Test (DFT)

↓

Physical Design

(Floor planning → Power Planning → Placement

→ Clock Tree Synthesis (CTS) → Routing

→ Static Timing Analysis (STA))

↓

Physical Verification (DRC/LVS)

↓

GDSII Generation (Tape-out)

↓

Fabrication

↓

Packaging and Testing

1. Requirements / Specification

The requirements of the chip are defined, including what the chip should do and the expected performance.

2. Architecture Design

The overall structure of the chip is planned by dividing it into different functional blocks.

3. RTL Design (Verilog/VHDL)

The chip is described using Hardware Description Languages (HDLs) such as Verilog or VHDL.

4. Functional Verification

The RTL design is simulated to verify that it works according to the specifications.

5. Logic Synthesis

The RTL code is converted into a gate-level netlist using synthesis tools.

6. Design for Test (DFT)

Extra test features are added to the design so the manufactured chip can be tested easily.

7. Physical Design

The gate-level netlist is converted into a physical chip layout. This stage includes:

- **Floor planning:** Decide the positions of major blocks.
- **Power Planning:** Design the power and ground network.
- **Placement:** Place the standard cells on the chip.
- **Clock Tree Synthesis (CTS):** Distribute the clock signal evenly.
- **Routing:** Connect all the cells using metal wires.
- **Static Timing Analysis (STA):** Check whether all timing requirements are met.

8. Physical Verification (DRC/LVS)

The layout is checked for design rule violations and to ensure it matches the original circuit.

9. GDSII Generation (Tape-out)

The final layout file (GDSII) is generated and sent to the semiconductor fabrication plant.

10. Fabrication

The chip is manufactured on a silicon wafer using semiconductor fabrication processes.

11. Packaging

The fabricated chip is placed inside a protective package and connected to external pins.

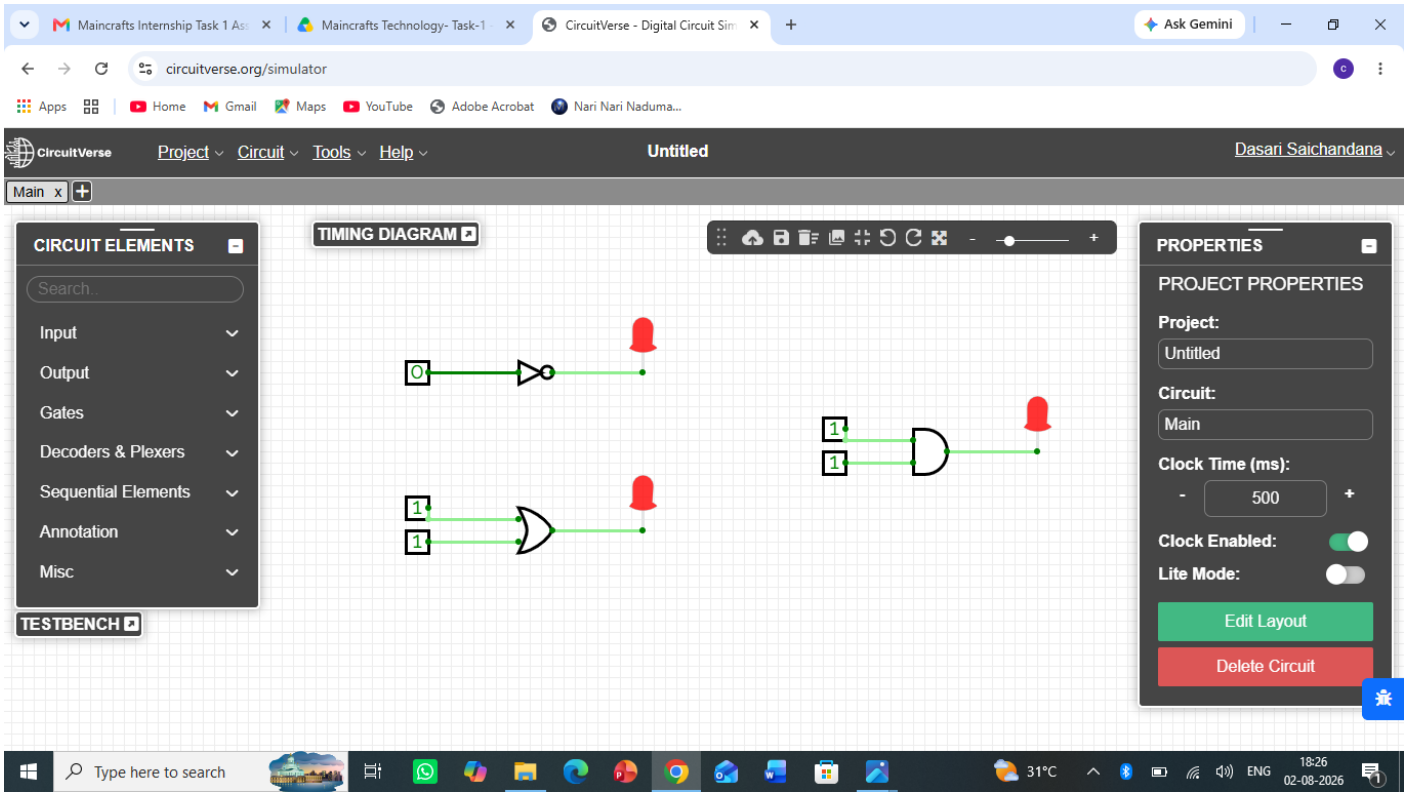
12. Testing

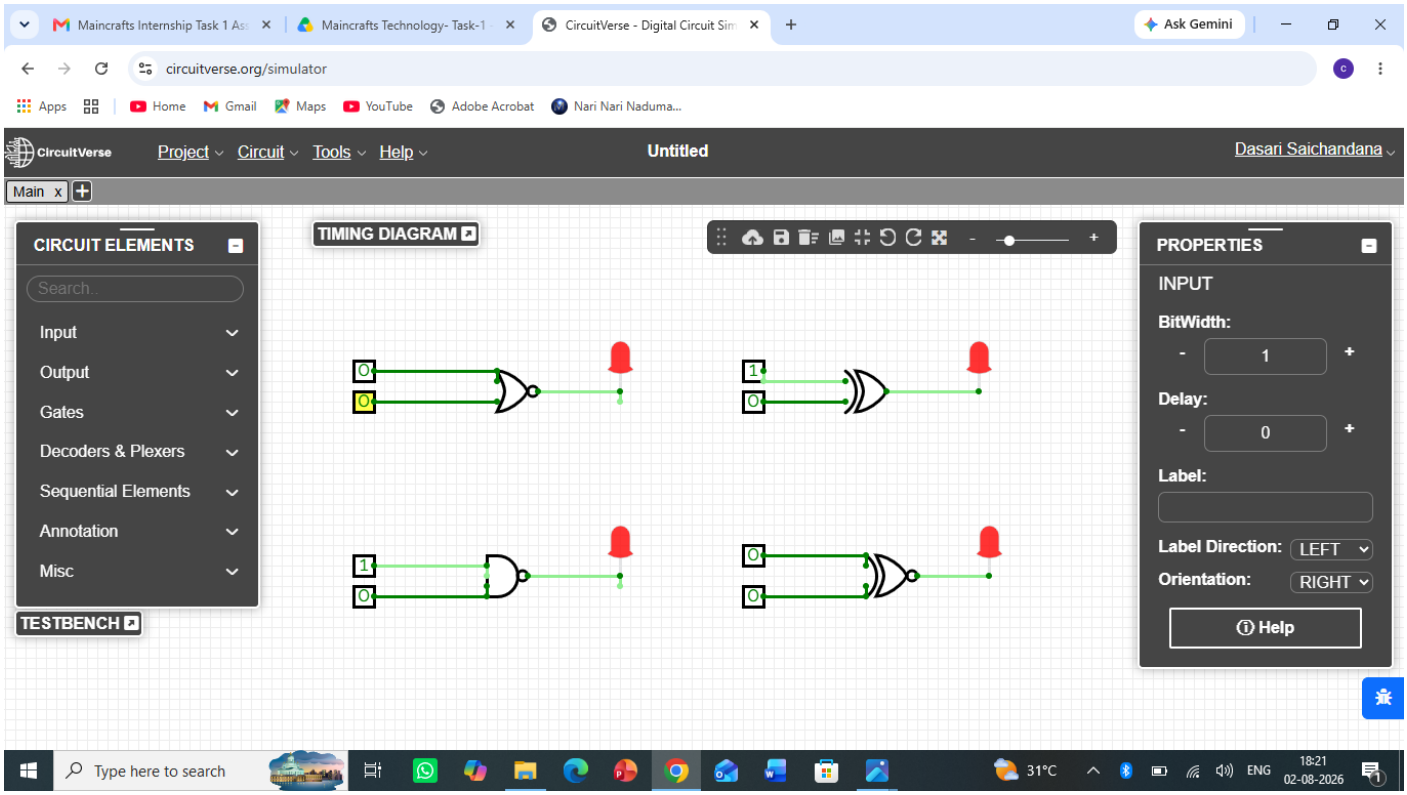
The packaged chip is tested to verify that it functions correctly before being released.

Implementation of Logic gate Simulations

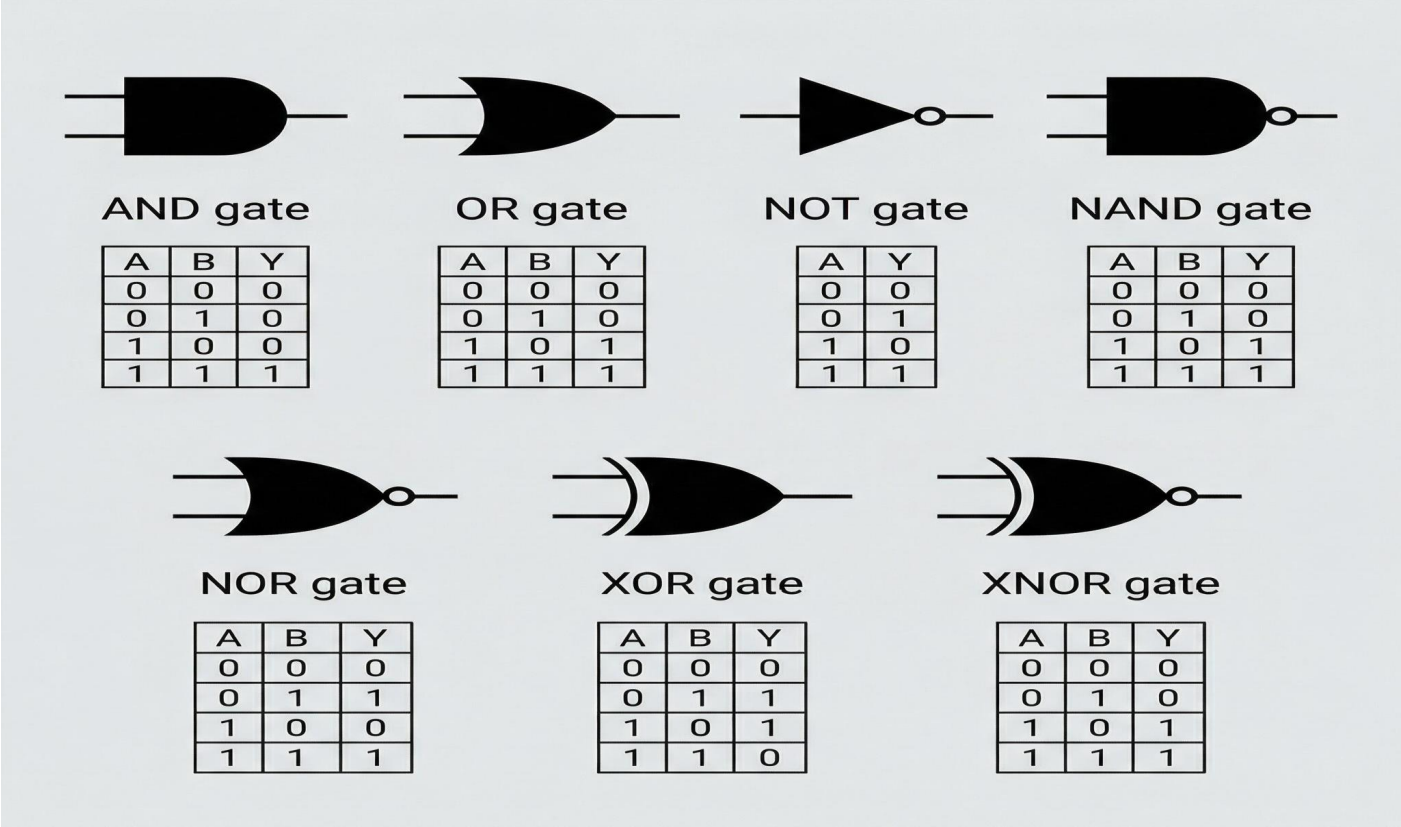
There are seven basic logic gates used in digital electronics: AND, OR, NOT, NAND, NOR, XOR, and XNOR. These logic gates are the fundamental building blocks of digital circuits. Each logic gate performs a specific logical operation based on the input values and produces the corresponding output. The AND, OR, NOT, and XOR gates were implemented and simulated using CircuitVerse, and their outputs were verified using truth tables.

The Simulated outputs are:

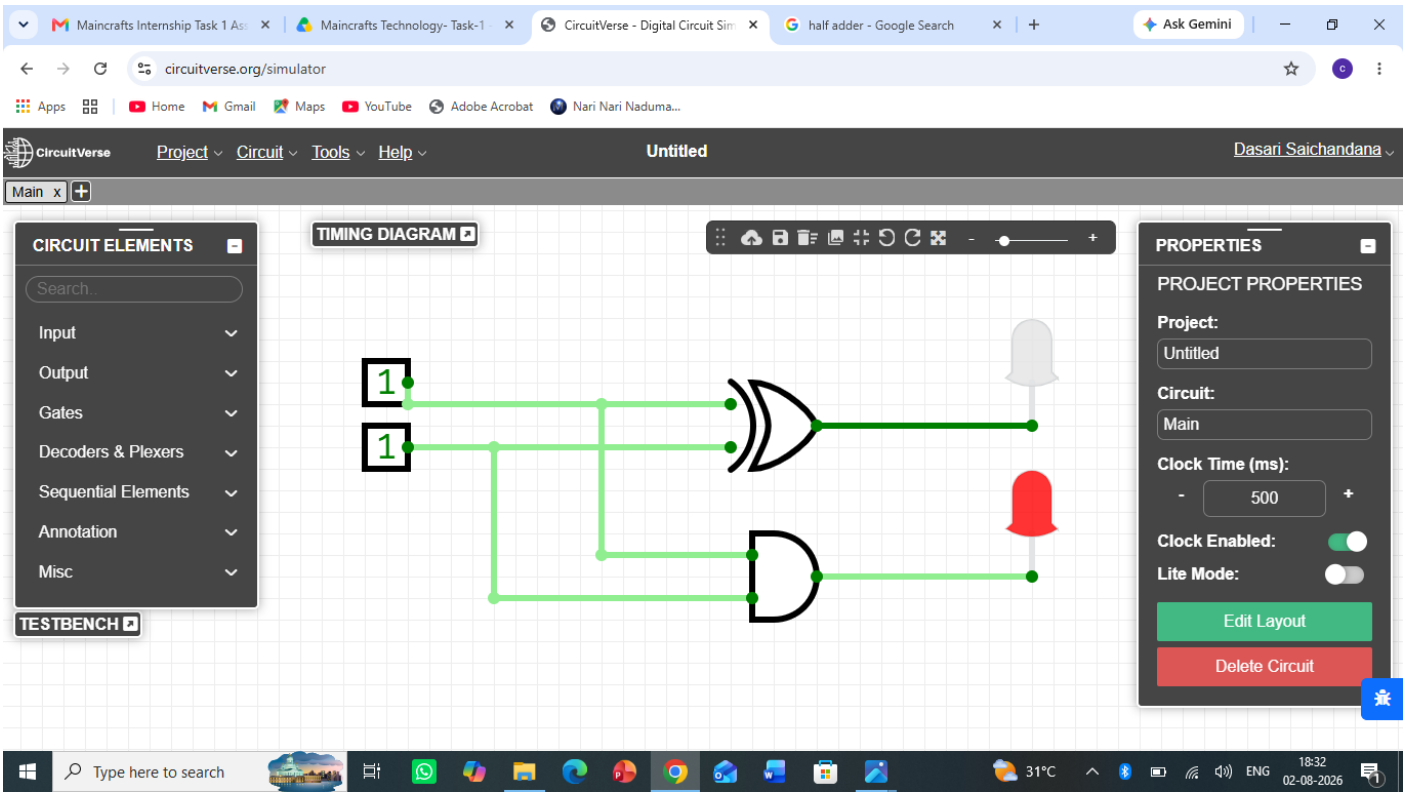


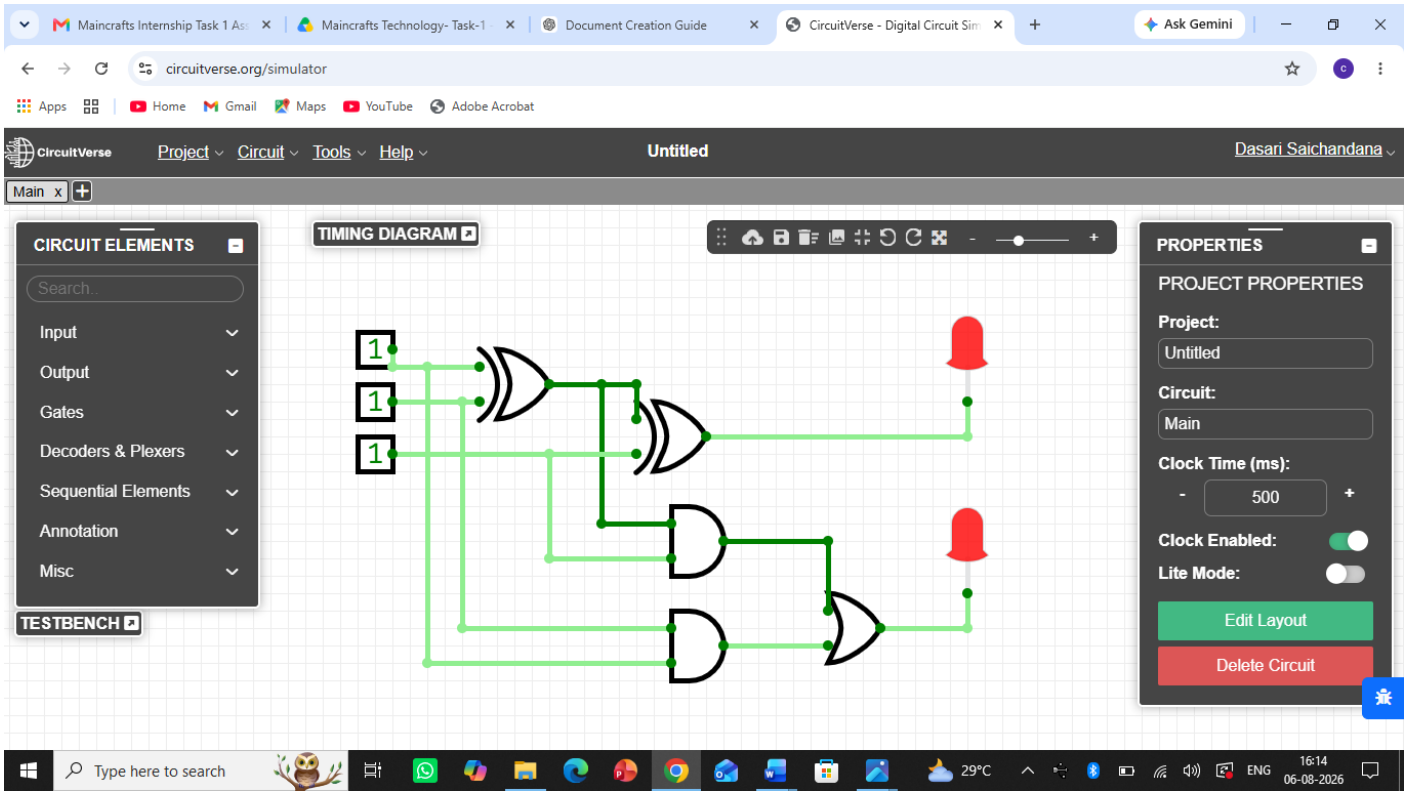


The Truth tables of the Logic gates



Simulation of Half adder and Full adder





Half Adder

A Half Adder is a combinational logic circuit that adds two single-bit binary inputs. It produces two outputs: Sum and Carry. The Sum is generated using an XOR gate, and the Carry is generated using an AND gate.

Full Adder

A Full Adder is a combinational logic circuit that adds three single-bit inputs: A, B, and Carry-in (Cin). It produces two outputs: Sum and Carry-out (Cout). A Full Adder can be implemented using two Half Adders and one OR gate.

Truth tables

HALF ADDER TRUTH TABLE

A	B	SUM (S)	CARRY (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

FULL ADDER TRUTH TABLE

A	B	CIN	SUM (S)	COUT
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Observation

The logic gates, Half Adder, and Full Adder were successfully implemented and simulated. The observed outputs matched the expected truth tables for all input combinations. Hence, the functionality of all the circuits was verified successfully.